Tax Deferral Burden and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Tax Deferral Burden (TDB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on TDB achieves an annualized gross (net) Sharpe ratio of 0.55 (0.50), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 48 (46) bps/month with a t-statistic of 4.65 (4.48), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Cash to assets, Equity Duration, Cash flow to market, Book to market using December ME, Market leverage, Operating Cash flows to price) is 29 bps/month with a t-statistic of 2.90.

1 Introduction

Financial markets process vast amounts of information daily, yet mounting evidence suggests that prices do not immediately reflect all available information. This deviation from the strong-form efficient market hypothesis creates predictable patterns in asset returns, as investors gradually incorporate complex accounting signals into their valuation models. While the literature has identified numerous accounting-based predictors of cross-sectional returns, the mechanisms through which intricate tax-related information affects price formation remain understudied.

We examine how Tax Deferral Burden (TDB), a measure capturing firms' accumulated tax obligations from timing differences between book and tax accounting, predicts future stock returns. Tax deferral information resides in financial statement footnotes and requires sophisticated analysis to interpret its implications for firm value. This complexity creates an ideal setting to test whether limited investor attention and gradual information diffusion generate predictable return patterns. Our analysis reveals that a value-weighted long-short strategy based on TDB generates significant abnormal returns, suggesting that markets slowly incorporate this tax-related information into prices.

The slow diffusion of information hypothesis provides a compelling framework for understanding why TDB predicts returns. When information is complex or obscured in financial statements, investors require time to process and incorporate it into their valuation models Hong and Stein (1999). Tax deferral burden represents precisely such information—it requires investors to parse detailed tax footnotes, understand the reversibility of timing differences, and assess implications for future cash flows. This processing complexity leads to systematic underreaction to TDB information, creating predictable return continuation patterns DellaVigna and Pollet (2009).

Limited investor attention amplifies the slow incorporation of TDB information into prices. Hirshleifer and Teoh (2003) demonstrate that investors have finite cognitive resources and must allocate attention across multiple information sources. Complex accounting signals like TDB receive less immediate attention than salient headline numbers, leading to gradual price adjustment as information diffuses through the market. The dispersion of information among investors further slows the incorporation process, as documented by Hong et al. (2016), who show that geographic dispersion of information leads to slower price discovery.

We predict that TDB's return predictability concentrates among stocks where information frictions are most severe. Small firms with low analyst coverage face greater information processing constraints, as fewer market participants scrutinize their financial statements Hong et al. (2000). These stocks should exhibit stronger return continuation following TDB signals, as information diffuses more slowly through their thinner information environments. Additionally, firms with more complex tax situations, reflected in higher absolute TDB values, should experience slower information incorporation as investors require more time to parse the implications Cohen et al. (2012).

Our empirical analysis reveals that TDB strongly predicts cross-sectional stock returns. The value-weighted long-short TDB strategy achieves an annualized gross Sharpe ratio of 0.55 and generates monthly average abnormal returns of 48 basis points relative to the Fama-French five-factor model plus momentum (t-statistic = 4.65). This performance places TDB in the top 7% of documented anomalies in terms of gross Sharpe ratio, demonstrating its economic significance. The strategy's alpha remains robust at 29 basis points per month (t-statistic = 2.90) even after controlling for the six most closely related anomalies and the Fama-French six factors.

The predictive power of TDB persists across various portfolio construction methodologies and market segments. Among large-cap stocks exceeding the 80th NYSE size percentile, the TDB strategy delivers average returns of 39 basis points per month (t-statistic = 3.05), with alphas ranging from 33 to 52 basis points across standard factor models. This large-cap performance is particularly noteworthy, as it demonstrates that TDB's predictive power extends beyond illiquid small stocks where many anomalies concentrate. After accounting for transaction costs using the composite bid-ask spreads from Chen and Velikov (2022), the net Sharpe ratio remains economically significant at 0.50.

TDB exhibits low correlations with existing anomalies and adds substantial value to the investment opportunity set. The highest correlation with any existing anomaly is modest, and TDB maintains significant alphas when controlling for closely related predictors including cash-to-assets, equity duration, and book-to-market ratios. When added to a combination strategy utilizing 158 existing anomalies, TDB increases terminal wealth by 31%, from $46.04 to $60.34 per dollar invested. These results confirm that TDB captures a distinct source of mispricing not spanned by the existing factor zoo.

This paper makes several contributions to the asset pricing and accounting literatures. First, we document a novel predictor of cross-sectional returns based on tax deferral burden, extending the literature on accounting-based anomalies. While Thomas and Zhang (2011) examine tax expense surprises and Lev and Nissim (2004) study the information content of taxable income, our focus on accumulated tax deferrals provides a distinct signal that captures long-term tax obligations rather than flow measures. Our results complement Balakrishnan et al. (2010), who document that tax reporting complexity affects information asymmetry, by showing that this complexity generates predictable return patterns through gradual information diffusion.

Methodologically, we employ the comprehensive testing framework of Novy-Marx and Velikov (2023) to establish the robustness of TDB's return predictability. This rigorous approach addresses concerns about data mining and p-hacking that plague the anomalies literature, as highlighted by Harvey et al. (2016). By examining TDB's performance across multiple portfolio construction methods, controlling for transaction costs, and testing its incremental contribution beyond 212 existing anomalies, we provide compelling evidence that TDB represents a genuine source of mispricing rather than a statistical artifact. Our analysis of net returns using high-frequency effective spreads ensures that the documented profits are achievable by real-world investors.

Our findings have important implications for market efficiency and the role of accounting information in price formation. The persistence of TDB's predictive power, especially among large-cap stocks with substantial institutional ownership, challenges the notion that sophisticated investors quickly eliminate mispricing of publicly available information. This evidence supports models of limited attention and gradual information diffusion, suggesting that even in modern markets with advanced information technology, complex accounting signals diffuse slowly through prices. For practitioners, our results highlight the value of analyzing detailed tax disclosures, while for academics, they underscore the importance of understanding how information complexity affects the speed of price discovery.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. We focus on U.S.-incorporated firms trading on major exchanges and exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) following standard practice in the literature.Tax Deferral Burden signal requires two key variables from firms' annual financial statements. Deferred taxes federal (TXDFED) represents the federal portion of deferred tax expense or benefit recognized in the income statement during the fiscal year. This item captures timing differences between book and tax accounting that will reverse in future periods, reflecting the tax consequences of temporary differences between financial reporting and tax bases of assets and liabilities. Earnings before interest and taxes (EBIT) measures a firm's operating performance before the effects of capital structure and tax considerations, calculated as revenues minus cost of goods sold and selling, general, and administrative expenses, including depreciation and amortization.construct the Tax Deferral Burden signal by dividing deferred taxes federal (TXDFED) by earnings before interest and taxes (EBIT) for each firm-year observation. This ratio captures the proportion of pre-tax operating earnings that represents deferred federal tax obligations, providing a scaled measure of the tax timing differences relative to the firm's operating performance. We calculate the signal using fiscal year-end values from annual financial statements and require non-missing values for both TXDFED and EBIT. To mitigate the influence of outliers, we exclude observations where EBIT equals zero or is negative, as these cases would produce economically meaningless or undefined ratios. The resulting measure indicates the extent to which a firm's current tax obligations are deferred relative to its operating earnings, with higher values suggesting a greater tax deferral burden.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the TDB signal. Panel A plots the time-series of the mean, median, and interquartile range for TDB. On average, the cross-sectional mean (median) TDB is 0.00 (-0.00) over the 1985 to 2024 sample, where the starting date is determined by the availability of the input TDB data. The signal's interquartile range spans -0.09 to 0.06. Panel B of Figure 1 plots the time-series of the coverage of the TDB signal for the CRSP universe. On average, the TDB signal is available for 5.37% of CRSP names, which on average make up 5.90% of total market capitalization.

4 Does TDB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on TDB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high TDB portfolio and sells the low TDB portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short TDB strategy earns an average return of 0.37% per month with a t-statistic of 3.46. The annualized Sharpe ratio of the strategy is 0.55. The alphas range from 0.34% to 0.48% per month and have t-statistics exceeding 3.18 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.27, with a t-statistic of -5.92 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 391 stocks and an average market capitalization of at least $1,389 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 13 bps/month with a t-statistics of 2.19. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between -5-34bps/month. The lowest return, (-5 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.67. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the TDB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the TDB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and TDB, as well as average returns and alphas for long/short trading TDB strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the TDB strategy achieves an average return of 39 bps/month with a t-statistic of 3.05. Among these large cap stocks, the alphas for the TDB strategy relative to the five most common factor models range from 33 to 52 bps/month with t-statistics between 2.54 and 4.32.

5 How does TDB perform relative to the zoo?

Figure 2 puts the performance of TDB in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the TDB strategy falls in the distribution. The TDB strategy's gross (net) Sharpe ratio of 0.55 (0.50) is greater than 93% (99%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the TDB strategy (red line).[2] Ignoring trading costs, a \$1 invested in the TDB strategy would have yielded \$4.24 which ranks the TDB strategy in the top 2% across the

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

212 anomalies. Accounting for trading costs, a \$1 invested in the TDB strategy would have yielded \$3.55 which ranks the TDB strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the TDB relative to those. Panel A shows that the TDB strategy gross alphas fall between the 66 and 90 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198506 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The TDB strategy has a positive net generalized alpha for five out of the five factor models. In these cases TDB ranks between the 83 and 99 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does TDB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of TDB with 209 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price TDB or at least to weaken the power TDB has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of TDB conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TDB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TDB} TDB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TDB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on TDB. Stocks are finally grouped into five TDB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TDB trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on TDB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the TDB signal in these Fama-MacBeth regressions exceed 0.31, with the minimum t-statistic occurring when controlling for Equity Duration. Controlling for all six closely related anomalies, the t-statistic on TDB is 0.77.

stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

Similarly, Table 5 reports results from spanning tests that regress returns to the TDB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the TDB strategy earns alphas that range from 28-51bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.84, which is achieved when controlling for Equity Duration. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the TDB trading strategy achieves an alpha of 29bps/month with a t-statistic of 2.90.

7 Does TDB add relative to the whole zoo?

Finally, we can ask how much adding TDB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the TDB signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 158-anomaly combination strategy grows to $46.04, while $1 investment in the combination strategy that includes TDB grows to $60.34.

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which TDB is available.

8 Conclusion

This study provides robust evidence that Tax Deferral Burden (TDB) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that TDB-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.55 gross (0.50 net), with monthly alphas of 48 basis points relative to the Fama-French five-factor model plus momentum. Notably, the signal maintains economic significance even when confronted with twelve competing factors, including closely related measures such as cash-to-assets and operating cash flows-to-price, yielding a monthly alpha of 29 basis points with robust statistical significance (t-statistic of 2.90). These findings suggest that TDB captures a distinct dimension of firm valuation that is not fully explained by existing factor models or related accounting-based signals. The economic magnitude and statistical robustness of the TDB effect indicate that tax-related accounting information contains valuable pricing information that market participants may not fully incorporate into stock prices. From a practical perspective, the relatively modest degradation from gross to net returns (approximately 5 basis points per month) suggests that the strategy remains implementable after accounting for realistic transaction costs. However, several limitations warrant consideration. First, the effectiveness of TDB may vary across different market regimes and regulatory environments, particularly given potential changes in tax legislation. Second, as with many accounting-based signals, data quality and availability may pose challenges, especially for smaller firms or in international markets. Future research should explore the underlying economic mechanisms driving the TDB effect, potentially examining whether it reflects systematic risk exposure, behavioral biases in processing tax information, or limits to arbitrage. Additionally, investigating the signal's performance in international markets and its interaction with tax regime changes could provide valuable insights into its robustness and economic foundations. Researchers might also benefit from examining the time-varying nature of the TDB premium and its relationship with macroeconomic conditions or tax policy uncertainty.

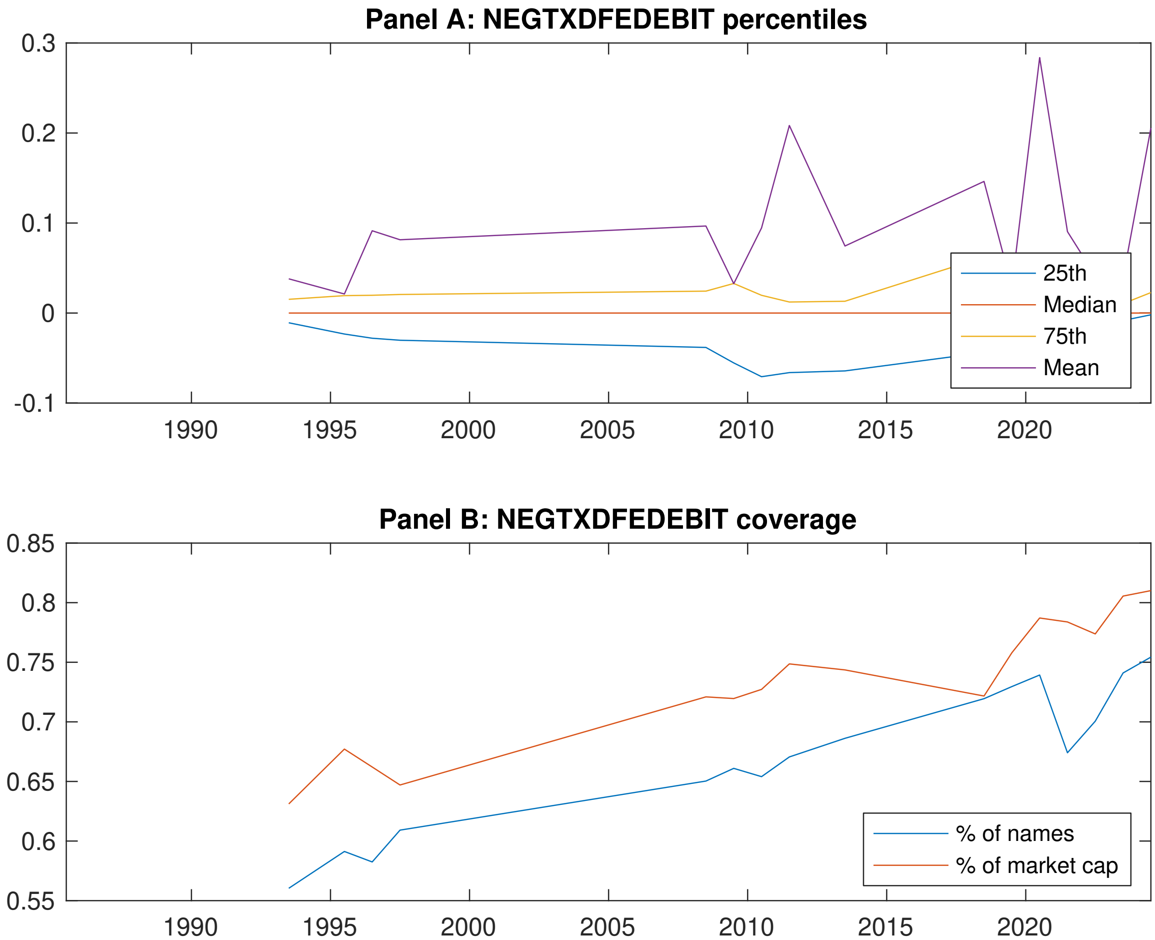


Figure 1: Times series of TDB percentiles and coverage.
This figure plots descriptive statistics for TDB. Panel A shows cross-sectional percentiles of TDB over the sample. Panel B plots the monthly coverage of TDB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on TDB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198506 to 202406.

Panel A: Excess returns and alphas on TDB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.56	0.82	0.70	0.73	0.93	0.37
	[2.37]	[3.95]	[3.21]	[3.07]	[3.81]	[3.46]
α_{CAPM}	-0.25	0.11	-0.05	-0.09	0.10	0.34
	[-3.14]	[1.61]	[-0.62]	[-1.42]	[1.18]	[3.18]
α_{FF3}	-0.27	0.09	-0.02	-0.07	0.14	0.41
	[-3.57]	[1.32]	[-0.26]	[-1.04]	[1.79]	[4.05]
α_{FF4}	-0.23	0.13	-0.00	-0.07	0.17	0.41
	[-3.10]	[1.88]	[-0.06]	[-1.04]	[2.18]	[3.98]
α_{FF5}	-0.27	-0.02	-0.07	-0.04	0.21	0.48
	[-3.49]	[-0.33]	[-1.00]	[-0.56]	[2.72]	[4.71]
α_{FF6}	-0.24	0.01	-0.06	-0.04	0.24	0.48
	[-3.13]	[0.21]	[-0.82]	[-0.57]	[3.06]	[4.65]
Panel B: Fama and French (2018) 6-factor model loadings for TDB-sorted portfolios						
β_{MKT}	1.07	0.97	0.98	1.09	1.07	-0.00
	[58.44]	[61.65]	[57.10]	[69.41]	[58.00]	[-0.11]
β_{SMB}	0.11	-0.01	0.15	-0.06	0.02	-0.09
	[3.87]	[-0.39]	[5.66]	[-2.42]	[0.61]	[-2.44]
β_{HML}	0.08	-0.05	-0.13	-0.13	-0.19	-0.27
	[2.29]	[-1.74]	[-4.00]	[-4.53]	[-5.58]	[-5.92]
β_{RMW}	-0.03	0.23	0.15	-0.10	-0.25	-0.22
	[-0.86]	[7.71]	[4.57]	[-3.27]	[-7.00]	[-4.62]
β_{CMA}	0.10	0.11	-0.02	0.04	0.17	0.07
	[2.09]	[2.55]	[-0.38]	[0.99]	[3.44]	[1.03]
β_{UMD}	-0.05	-0.07	-0.02	0.00	-0.05	0.00
	[-2.98]	[-4.39]	[-1.41]	[0.13]	[-2.73]	[0.17]
Panel C: Average number of firms (n) and market capitalization (me)						
n	489	391	910	703	508	
me ($\$10^6$)	1389	2381	2463	2846	2480	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the TDB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198506 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.37	0.34	0.41	0.41	0.48	0.48
			[3.46]	[3.18]	[4.05]	[3.98]	[4.71]	[4.65]
Quintile	NYSE	EW	0.13	0.10	0.12	0.11	0.11	0.11
			[2.19]	[1.72]	[2.09]	[1.94]	[1.90]	[1.80]
Quintile	Name	VW	0.26	0.21	0.27	0.27	0.37	0.36
			[2.44]	[1.98]	[2.80]	[2.70]	[3.75]	[3.64]
Quintile	Cap	VW	0.34	0.28	0.35	0.34	0.47	0.46
			[2.98]	[2.47]	[3.36]	[3.23]	[4.59]	[4.45]
Decile	NYSE	VW	0.38	0.38	0.45	0.40	0.53	0.49
			[2.58]	[2.57]	[3.15]	[2.79]	[3.66]	[3.37]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.34	0.32	0.37	0.37	0.46	0.46
			[3.17]	[2.98]	[3.72]	[3.69]	[4.55]	[4.48]
Quintile	NYSE	EW	-0.05					
			[-0.67]					
Quintile	Name	VW	0.23	0.19	0.24	0.24	0.35	0.34
			[2.18]	[1.80]	[2.49]	[2.44]	[3.56]	[3.50]
Quintile	Cap	VW	0.31	0.25	0.31	0.30	0.43	0.43
			[2.74]	[2.17]	[2.91]	[2.85]	[4.24]	[4.14]
Decile	NYSE	VW	0.34	0.35	0.40	0.37	0.50	0.47
			[2.31]	[2.33]	[2.82]	[2.63]	[3.44]	[3.28]

Table 3: Conditional sort on size and TDB

This table presents results for conditional double sorts on size and TDB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on TDB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high TDB and short stocks with low TDB .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198506 to 202406.

Panel A: portfolio average returns and time-series regression results

	TDB Quintiles					TDB Strategies					
Size quintiles	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.82	0.55	0.20	0.55	0.98	0.16	0.15	0.17	0.14	0.08	0.06
	[2.81]	[1.75]	[0.49]	[1.50]	[3.13]	[1.30]	[1.19]	[1.31]	[1.05]	[0.62]	[0.49]
(2)	0.74	0.87	0.39	0.60	0.96	0.22	0.19	0.22	0.21	0.25	0.24
	[2.64]	[3.20]	[1.04]	[1.99]	[3.23]	[2.27]	[1.95]	[2.39]	[2.26]	[2.61]	[2.51]
(3)	0.76	0.78	0.65	0.81	0.84	0.08	0.07	0.12	0.10	0.15	0.14
	[2.70]	[3.09]	[1.95]	[2.86]	[3.00]	[0.70]	[0.56]	[1.04]	[0.92]	[1.29]	[1.18]
(4)	0.89	0.78	0.82	0.85	0.94	0.05	0.00	0.08	0.04	0.22	0.18
	[3.43]	[3.27]	[2.88]	[3.20]	[3.39]	[0.39]	[0.01]	[0.70]	[0.35]	[1.82]	[1.51]
(5)	0.54	0.81	0.75	0.75	0.92	0.39	0.33	0.39	0.39	0.52	0.52
	[2.36]	[3.98]	[3.57]	[3.30]	[3.77]	[3.05]	[2.54]	[3.20]	[3.16]	[4.32]	[4.26]

Panel B: Portfolio average number of firms and market capitalization

	TDB Quintiles: Average n					TDB Quintiles: Average market capitalization ($\$10^6$)				
Size quintiles	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
(1)	345	341	336	337	346	41	37	29	32	42
(2)	97	97	97	97	96	66	68	64	65	66
(3)	63	64	64	64	63	110	111	109	112	110
(4)	51	51	51	52	51	234	241	236	239	239
(5)	48	48	48	48	48	1503	1753	1993	2008	2052

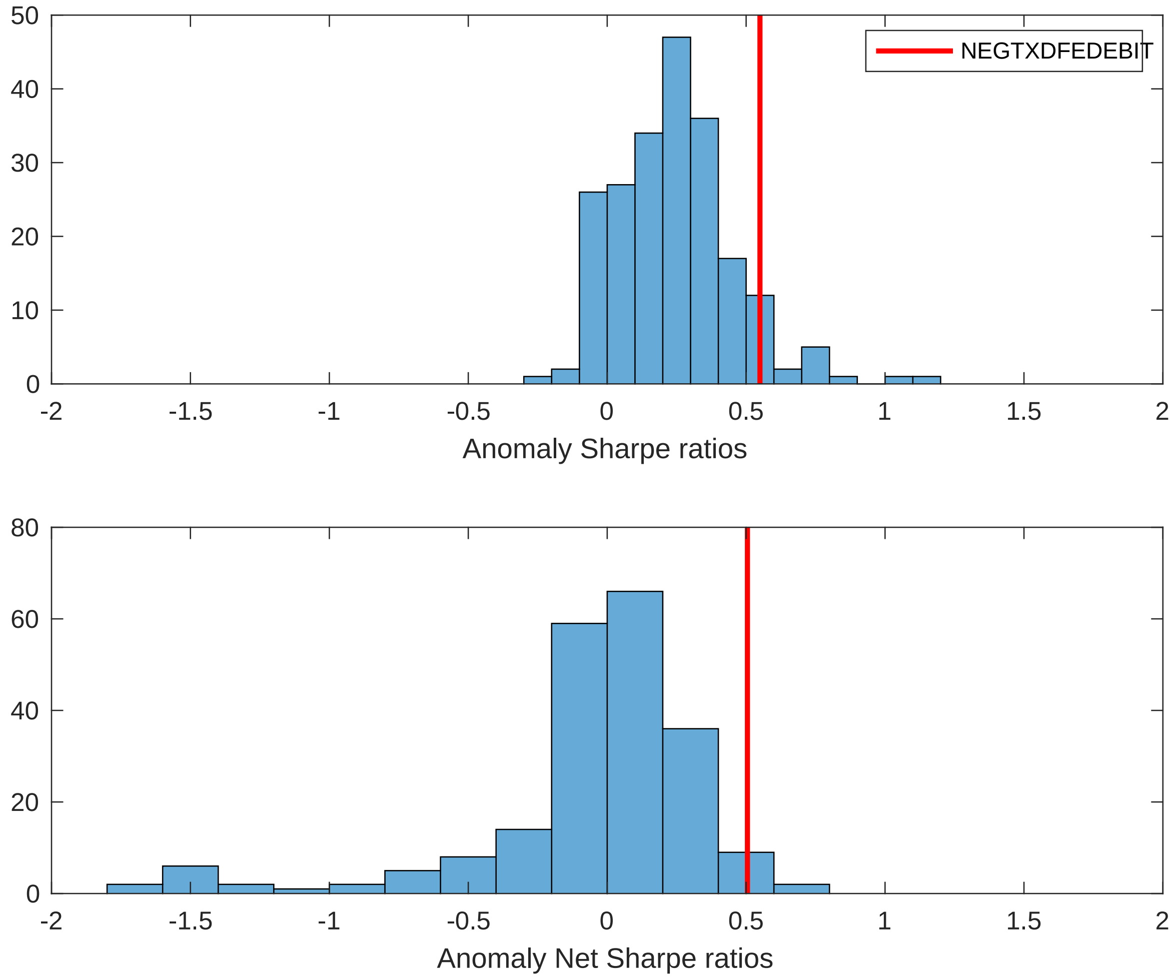


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the TDB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

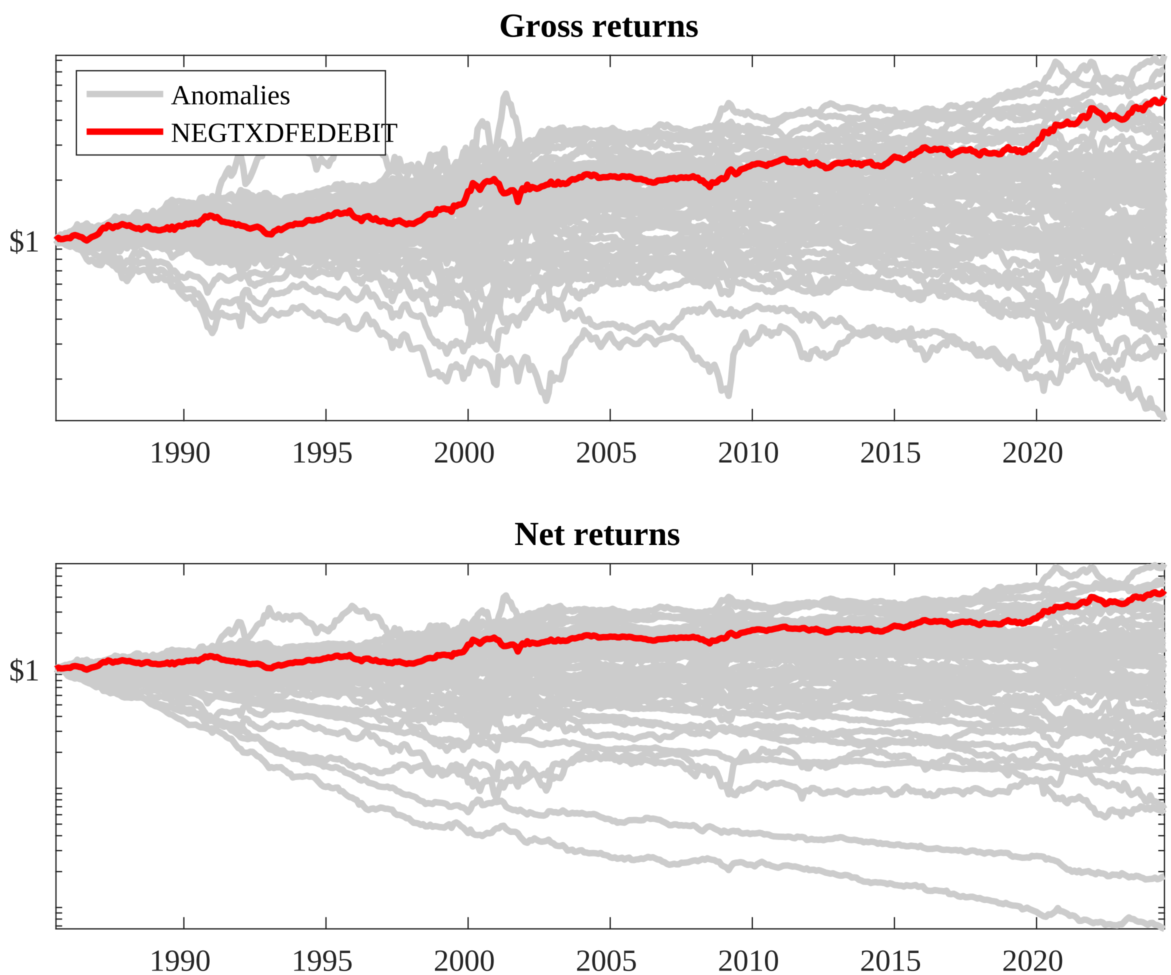
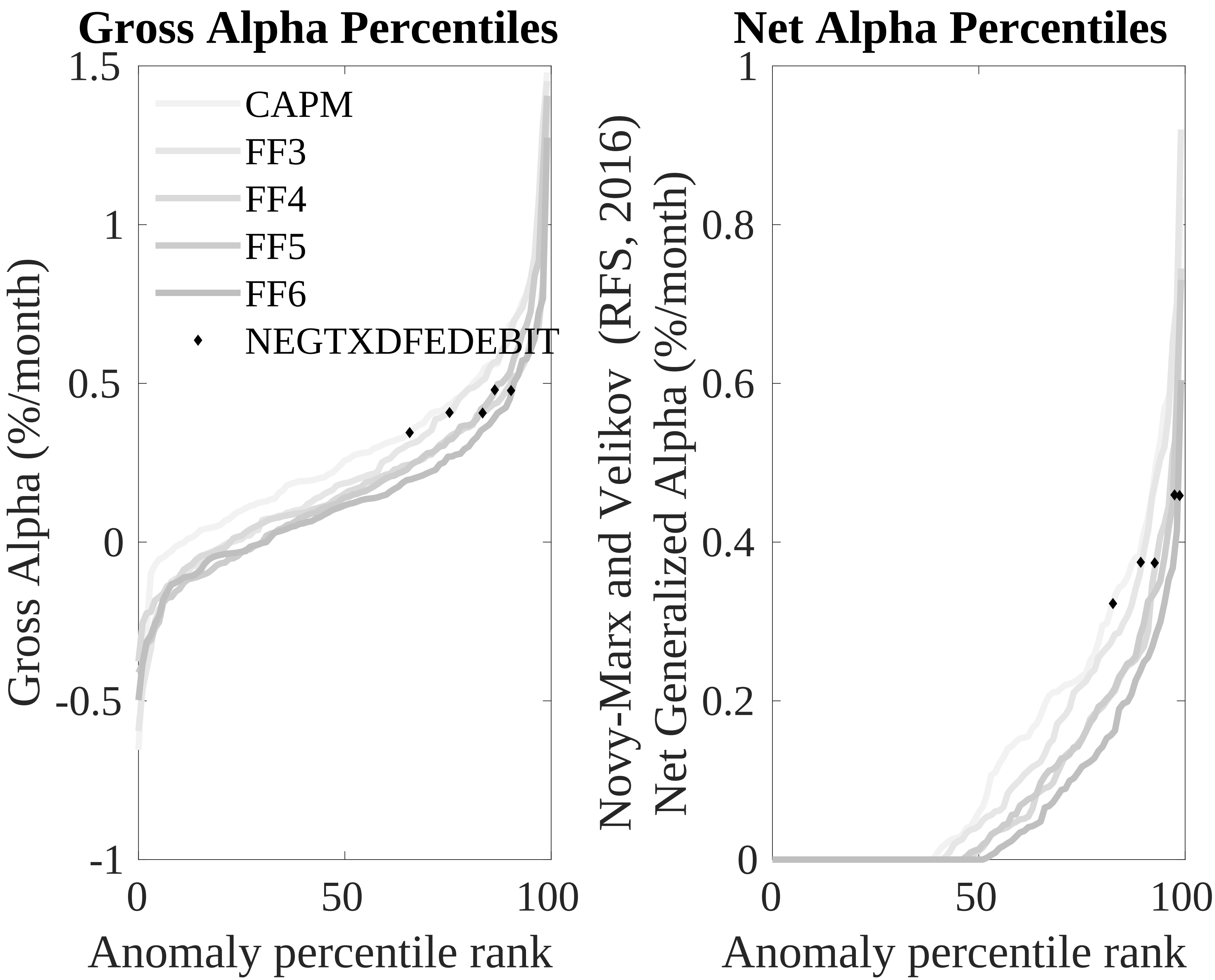


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the TDB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGTXDFEDEBIT
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100

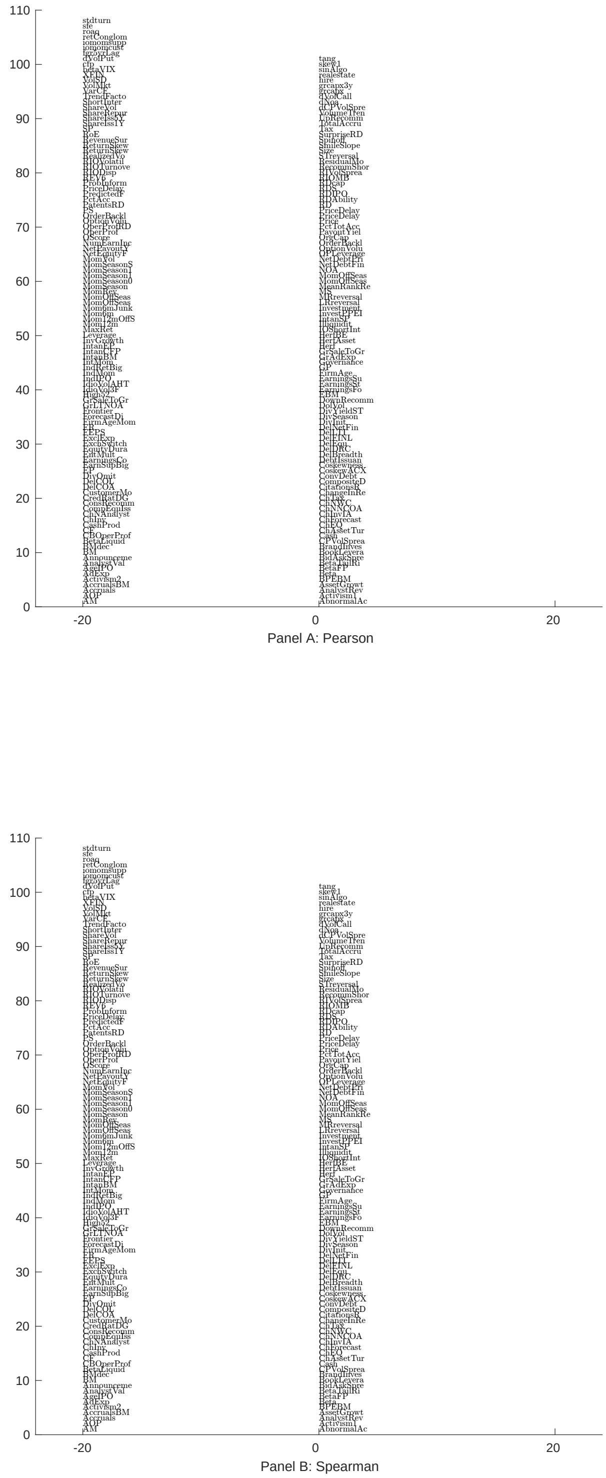


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 209 filtered anomaly signals with TDB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

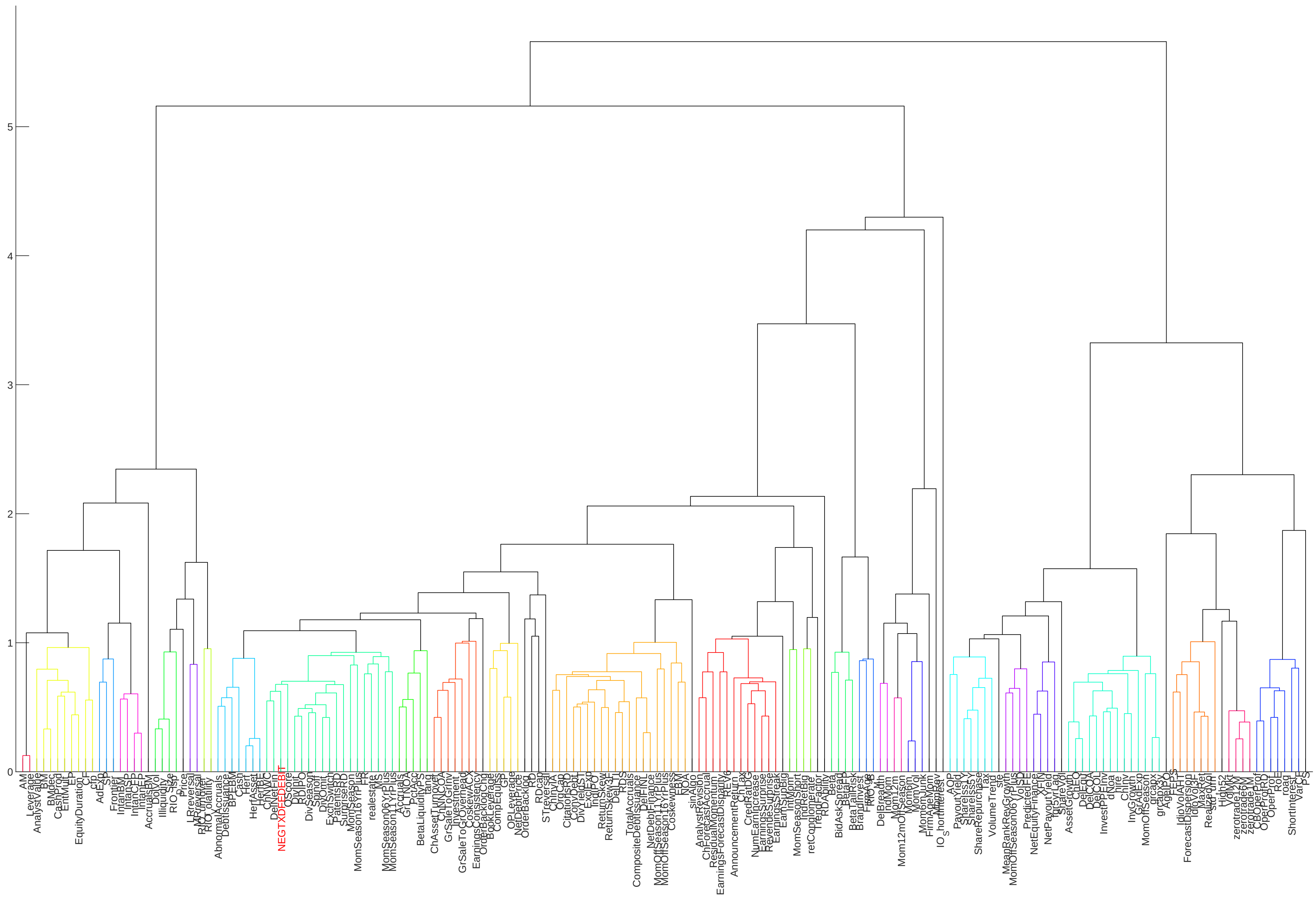


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

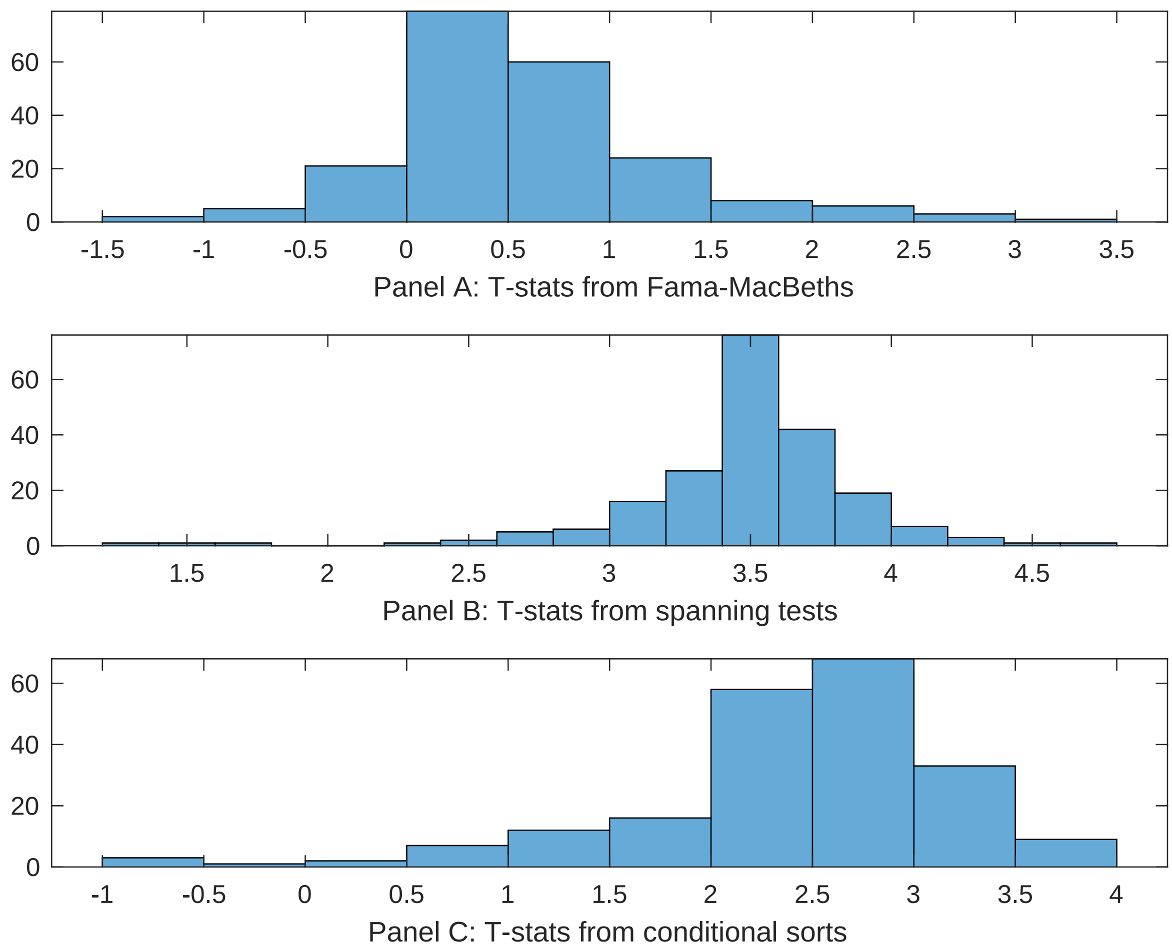


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of TDB conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TDB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TDB} TDB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TDB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on TDB. Stocks are finally grouped into five TDB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TDB trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on TDB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{TDB}TDB_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k}X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Cash to assets, Equity Duration, Cash flow to market, Book to market using December ME, Market leverage, Operating Cash flows to price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198506 to 202406.

Intercept	0.10	0.16	0.10	0.93	0.10	0.10	0.12
	[3.69]	[6.32]	[3.50]	[3.04]	[3.51]	[3.46]	[4.81]
TDB	0.78	0.18	0.24	0.34	0.39	0.34	0.43
	[1.35]	[0.31]	[0.43]	[0.59]	[0.69]	[0.59]	[0.77]
Anomaly 1	0.62						0.11
	[1.56]						[3.13]
Anomaly 2		0.25					0.23
		[2.66]					[3.41]
Anomaly 3			0.18				-0.23
			[0.58]				[-0.98]
Anomaly 4				0.27			0.20
				[3.22]			[2.84]
Anomaly 5					0.13		-0.25
					[0.61]		[-1.32]
Anomaly 6						0.85	0.79
						[2.95]	[4.36]
# months	463	468	463	463	463	463	463
$\bar{R}^2(\%)$	1	0	1	0	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the TDB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{TDB} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Cash to assets, Equity Duration, Cash flow to market, Book to market using December ME, Market leverage, Operating Cash flows to price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198506 to 202406.

Intercept	0.28	0.47	0.48	0.46	0.47	0.51	0.29
	[2.84]	[4.63]	[4.73]	[4.56]	[4.56]	[5.00]	[2.90]
Anomaly 1	33.90						30.37
	[8.54]						[7.01]
Anomaly 2		-13.35					-1.87
		[-2.49]					[-0.29]
Anomaly 3			-8.48				2.14
			[-2.40]				[0.48]
Anomaly 4				-25.89			-9.98
				[-3.87]			[-1.45]
Anomaly 5					-12.58		-6.98
					[-3.00]		[-1.62]
Anomaly 6						-13.55	-3.78
						[-3.62]	[-0.87]
mkt	-6.58	-0.34	-0.21	-1.30	1.38	0.27	-5.33
	[-2.77]	[-0.14]	[-0.09]	[-0.54]	[0.56]	[0.11]	[-2.14]
smb	-6.87	-5.10	-7.16	-0.88	-7.09	-5.87	-3.35
	[-2.00]	[-1.33]	[-1.95]	[-0.22]	[-1.94]	[-1.60]	[-0.86]
hml	-12.23	-10.22	-19.96	1.74	-8.23	-14.92	9.45
	[-2.75]	[-1.36]	[-4.06]	[0.21]	[-1.16]	[-2.86]	[1.00]
rmw	-5.99	-22.11	-19.30	-29.19	-24.13	-17.73	-10.77
	[-1.26]	[-4.77]	[-4.01]	[-5.93]	[-5.18]	[-3.71]	[-1.97]
cma	18.37	1.37	8.59	6.95	3.20	8.92	15.48
	[2.92]	[0.20]	[1.29]	[1.07]	[0.49]	[1.36]	[2.28]
umd	0.42	0.48	-2.98	1.58	-2.06	-4.75	-1.50
	[0.20]	[0.21]	[-1.07]	[0.69]	[-0.83]	[-1.73]	[-0.49]
# months	464	468	464	464	464	464	464
$\bar{R}^2(\%)$	29	19	18	20	19	19	29

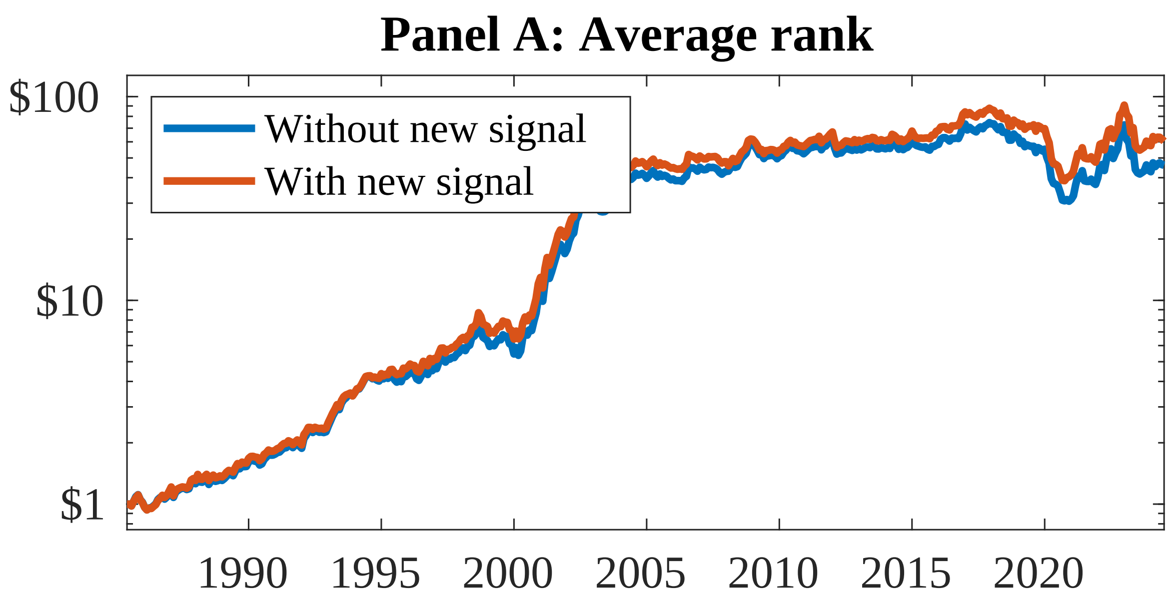


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as TDB. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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